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Survival assay of transiently transfected dopaminergic neurons

Li-ying Yu, Urmas Arumäe *

Institute of Biotechnology, University of Helsinki, FIN-00014, Helsinki, Finland

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Abstract

Death pathways in the apoptotic neurons are mostly studied by manipulating the levels of apoptosis-related proteins and counting the survival/death of affected neurons. Such assays are, however, technically complicated. We developed a transfection–survival assay for cultured embryonic dopaminergic (DA) neurons induced to die by deprivation of glial cell line-derived neurotrophic factor (GDNF). The calcium phosphate co-precipitation technique was used to transfect DA neurons. Microisland cultures and co-transfected enhanced green fluorescent protein allowed direct counting of transfected neurons from the same cultures at the beginning and the end of GDNF deprivation, whereas post hoc subtraction of tyrosine hydroxylase-negative neurons allowed exclusion of transfected non-DA neurons. Overexpression of dominant-negative mutant of caspase-6 significantly blocked the death of GDNF-deprived DA neurons. Thus, we have found a tool not only to transfect the neurons dissociated from midbrain, but also to analyze the apoptotic proteins particularly in DA neurons.

Keywords: Dopaminergic neurons; GDNF; Caspase-6; Death pathways; Survival assay; Transfection

1. Introduction

Cell death is an organized process that can occur via different molecular–cellular pathways. Two main classical apoptotic pathways are known: extrinsic (death receptor) or intrinsic (mitochondrial) (Danial and Korsmeyer, 2004; Thorburn, 2004; Riedl and Salvesen, 2007). Both result in the activation of caspases that ultimately demolish the cell. There are also several non-apoptotic death pathways (Clarke, 1990; Jäättelä, 2002; Mehlen and Bredesen, 2004; Lockshin and Zakeri, 2004). Moreover, the caspases can also be activated by "non-classical" methods that are neither extrinsic nor intrinsic (Mehlen et al., 1998; Yu et al., 2003). This is often the case in pathological situations.

The neurons can die either physiologically, e.g. during ontogenetic death periods where target-derived neurotrophic factors suppress the intrinsic apoptotic program of the innervating neurons (Barde, 1989; Oppenheim, 1991; Johnson and Deckwerth, 1993; Chang et al., 2002; Krieglstein, 2006; Bredesen et al., 2006), or pathologically, e.g. in the neurodegenerative diseases and damage (Gibson, 2001; Vila and Przedborski, 2003; Fadeel and Orrenius, 2005). The death pathways can be considerably different in each case. Whereas most of the ontogenetic neuronal death occurs via classical mitochondrial apoptosis, the sick or damaged neurons can die via various "non-classical" or mixed pathways. The midbrain dopaminergic (DA) neurons undergo an ontogenetic death period postnatally (Jackson-Lewis et al., 2000). GDNF (glial cell line-derived neurotrophic factor) (Airaksinen and Saarma, 2002; Bespalov and Saarma, 2007) is one of the factors that promote the survival of DA neurons, both in vitro (Lin et al., 1993; Burke et al., 1998) and in vivo (Oo et al., 2003, 2005). DA neurons also degenerate in Parkinson's disease (Bergman and Deuschl, 2002; Jenner and Olanow, 2006). GDNF significantly improves the survival of DA neurons in the mouse models of Parkinson's disease (Airaksinen and Saarma, 2002; Bespalov and Saarma, 2007; Lindholm et al., 2007) and, in some studies, also in the Parkinson's patients (Gill et al., 2003; Slevin et al., 2005), although other studies reported no such effect (Lang et al., 2006). Thus, understanding the death pathways suppressed by GDNF (and activated by its absence) is important for not only basic neurobiology but also potential designing treatments for neurodegenerative diseases. However, the death pathways in the DA neurons are very poorly studied, in part due to technical difficulties.

* Corresponding author at: Biocentre 1, P.O. Box 56, Viikinkaari 9, FIN-00014, University of Helsinki, Finland. Tel.: +358 9 19159396; fax: +358 9 19159366.

E-mail address: urmas.arumae@helsinki.fi (U. Arumäe).

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Elucidation of the apoptotic pathways requires experimental manipulation of the cells. For example, the reductionistic approach in cultured neurons where the candidate proteins are overexpressed or suppressed, followed by estimation of the survival/death of the transfected neurons, has been informative. Cultured DA neurons have several intrinsic difficulties hampering such transfection–survival assay. First, a small number of neurons is transfected by the current techniques that could be enough for the assay, but not enough for "in batch" measurement of the effects. Thus, the transfected neurons must be counted directly under the microscope (this requires a marker). Second, the midbrain cultures are mixed, containing many other neuronal types that all get transfected. Thus, the transfected DA neurons must be recognized among transfected non-DA neurons (this requires a second marker). Third, the efficiency of transfection can vary in different dishes. Therefore, the survival/death of the transfected neurons should be followed in each dish individually, at the beginning and at the end of the death stimulus.

We elaborated the transfection–survival assay for the embryonic DA neurons in the dissociated midbrain cultures where the DA neurons are induced to die by removal of neurotrophic support–deprivation of GDNF. We found that usage of co-transfected plasmid for eGFP (enhanced green fluorescent protein) and post hoc immunostaining for tyrosine hydroxylase (TH), a specific marker for DA neurons, solves the above-mentioned problems, so that the effects of apoptosis-related proteins, e.g. caspase-6 in the death pathways triggered by GDNF removal, can be reliably studied.

2. Materials and methods

2.1. Cultures of E13 mouse ventral mesencephalon

The midbrain floors from embryonic day (E) 13 embryos of NMRI strain mice were prepared under stereomicroscope. The tissues were digested with 0.5% trypsin (ICN Biomedical, Aurora, OH) in the Ca^{2+}-Mg^{2+}-free Hanks Balanced Salt Solution (CMF-HBSS) (Invitrogen/Gibco, Carlsbad, CA) for 15 min at 37 °C. The tissue pieces were triturated into single-cell suspension with siliconized autoclaved fire-polished glass pipettes in the CMF-HBSS containing 50% of fetal calf serum (Hyclone, Logan, UT) and 0.1 mg/ml of DNase I (Roche, Basel, Switzerland). Fetal calf serum was removed by washing the cells three times with complete culture medium: DMEM/F12 (Invitrogen/Gibco) containing N2 serum supplement (Invitrogen/Gibco), 33 mM D-glucose (Sigma-Aldrich, Louis, MO), and L-Glutamine (Invitrogen/Gibco). The cells were then plated onto the culture dishes coated with poly-ornithine (Sigma). To get the comparable cell number for different experimental points, only the microisland areas of standard diameter (3 mm), scratched to the dishes, were coated. Equal volumes of cell suspension (4 μl, containing 1000–3000 neurons) were plated onto these areas. The neurons were allowed to attach in DMEM/F12 containing the N2 supplement (Invitrogen/Gibco) for 1 h, then more medium was added including 100 ng/ml of recombinant human GDNF (PeproTech Inc., Rocky Hill, New Jersey). The cultures were grown with GDNF for 5–6 days in vitro (DIV).

2.2. Transfections and counting

The 6 DIV GDNF-maintained cultures were transfected with the expression plasmids of interest at 1 μg/μl together with the reporter plasmid for enhanced green fluorescent protein (eGFP) (Clontech, Mountain View, CA) at 0.2 μg/μl (ratio 5:1) using the Calcium Phosphate Transfection Kit (InVitrogen, #K2780-01) according to the manufacturers instructions. The cultures were grown overnight with GDNF and examined then for the number of eGFP-expressing neurons. Unsuccessfully transfected cultures were discarded. To remove GDNF, the cultures were gently washed with GDNF-free medium three times. In some experiments the dishes were washed in the same way, but GDNF was added back. The number of initial fluorescent neurons was then counted microscopically for every dish. Three days (70–75 h) later the cultures were fixed with fresh 4% paraformaldehyde in the phosphate-buffered saline (PBS), post-fixed with ice-cold acetone, blocked with 10% of horse serum (InVitrogen) in PBS and stained with sheep polyclonal antibodies to TH (Chemicon, Temecula, CA) at 1:200 dilution. The immunoreactivity was visualized with anti-sheep secondary antibodies conjugated to Cy3. The number of eGFP-TH double-positive neurons (final number) was counted from each dish. eGFP-positive but TH-negative neurons were counted separately and subtracted from the initial counts. The number of initial neurons was set to 100% and the survival rate of the neurons at three days of GDNF deprivation (final number) was expressed as a percent of initial neurons. Every experimental point was always set in three parallels, and the whole assay was repeated at least three times on independent cultures. The plasmids for dominant-negative caspase-6 (Yu et al., 2003) and FLAG-GFRα1 (Yang et al., 2004) are described; the plasmid for XIAP-Myc is a gift from Dan Lindholm, Minerva Institute, University of Helsinki.

2.3. Immunocytochemistry

E13 midbrain floor neurons were grown on poly-ornithine-coated plastic dishes or glass coverslips, fixed with 4% PFA in PBS, permeabilized with 0.3% Triton X-100 (Sigma-Fluka, Buchs, Switzerland) and then blocked with 10% horse serum (InVitrogen) in TBS containing 0.3% Triton X-100. The following antibodies were used: TH, FLAG (Sigma), Myc (InVitrogen) and the Cy3-conjugated secondary antibodies (Jackson ImmunoResearch Laboratories, Inc, West Grove, PA). The specimens were mounted in Vectashield (Vector Laboratories, Burlingame, CA). The images were captured at RT with an inverted microscope (model DM-IRB; Leica, Germany) using HC PL FLUOTAR objective (10×/0.30), and a 3CCD color video camera (model DXC-950P; Sony) under the control of Image-Pro Plus software version 3.0 (Media Cybernetics). Individual images were identically processed with Photoshop CS3 (Adobe Systems, San Jose, CA).

2.4. Immunoblotting

The midbrain cultures maintained for 5 days with GDNF were deprived of GDNF or maintained further with GDNF for 36 h, then lysed in the Laemmli sample buffer and analyzed by immunoblotting with antibodies recognizing large subunit of cleaved caspase-6 (Cell Signaling Technology, MA). To show equal loading of the proteins, the filter was reprobed with antibodies to actin (Sigma-Fluka, Switzerland).

3. Results

3.1. Culture conditions for GDNF-dependent embryonic dopaminergic neurons

To study the death pathways activated in the GDNF-deprived DA neurons we first established the dissociated culture conditions. E13 mouse midbrain floors were dissected and dissociated into single cell suspension. The cells were grown at high density as microislands on the standard small (3 mm diameter) areas (Fig. 1). The cultures were maintained with GDNF for 5–6 days. GDNF was then withdrawn causing significant death of the DA neurons due to the lack of the neurotrophic support. Characterization of this death will be described elsewhere. Fig. 1 shows our typical midbrain cultures grown with GDNF for 6 DIV, then deprived of GDNF or grown further with GDNF for three additional days. Equal number of neurons was plated in both cultures. To visualize DA neurons that constitute only a fraction of all neurons, the cultures are stained with antibodies to TH, a specific marker for DA neurons. GDNF deprivation causes robust reduction in the number of TH-positive neurons that is, however, not obvious from the phase-contrast image.

3.2. Transfection of the dopaminergic neurons

To transfect the DA neurons we tried several techniques. Reasonable transfection efficiency was obtained with the calcium phosphate co-precipitation technique. Still from time to time the experiments failed, resulting in total death of the cultures or a too low number of transfected neurons. Therefore, our practice was to maintain the transfected cultures overnight to estimate the transfection efficiency by co-transfected eGFP fluorescence. Only successfully transfected cultures containing a reasonable number of fluorescent neurons were chosen for the experiments. Maintaining the cultures overnight has an additional advantage of having the transfected proteins expressed already by the time of GDNF deprivation. Fig. 2 (Day 0) illustrates our typical culture imaged the next day after transfection with eGFP-encoding plasmid.

3.3. eGFP co-transfection

To visualize and count the transfected neurons in living cultures we co-expressed the plasmid for eGFP together with the plasmid of interest. Such an approach requires, however, that the eGFP-expressing neuron always expresses the co-transfected plasmid. To verify such co-expression we co-transfected eGFP with another reporter plasmid encoding for the proteins fused to either FLAG- or Myc-tag. The cultures were then immunostained with anti-FLAG or anti-Myc antibodies and visualized by red-emitting (Cy3-conjugated) secondary antibodies. We then counted the double-fluorescent (green and red) neurons. When the plasmid for eGFP was in short-

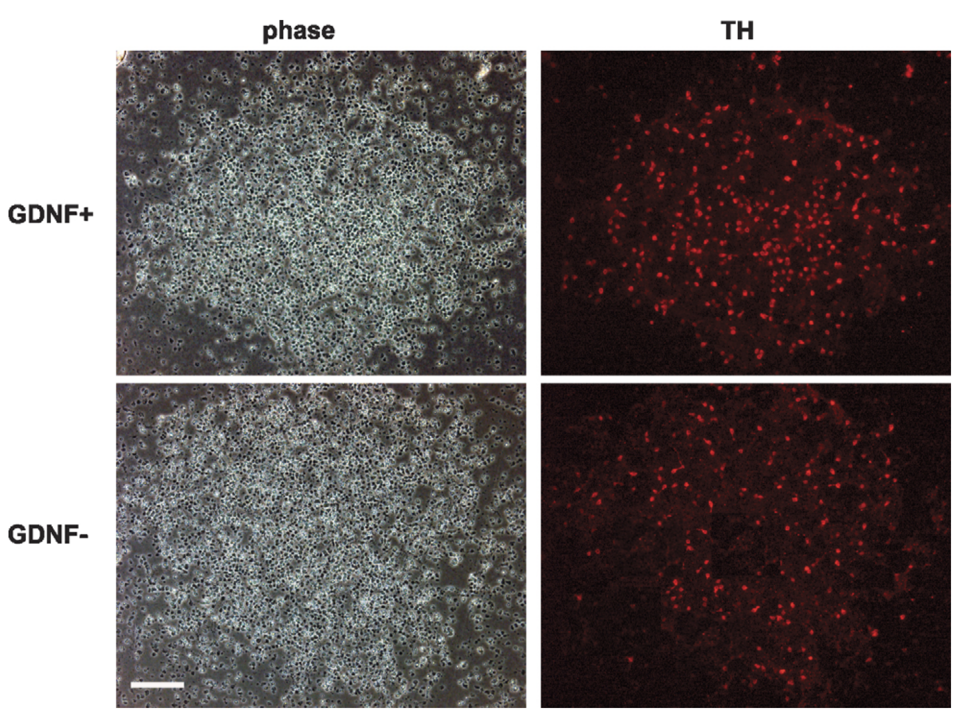


Fig. 1. GDNF deprivation dramatically reduces the number of dopaminergic neurons. Typical examples of 6 DIV GDNF-dependent mouse midbrain floor cultures maintained with GDNF (GDNF+) or deprived of GDNF (GDNF-) for 3 additional DIV. Dopaminergic neurons are visualized by TH immunostaining, whereas all cells in the same fields are visualized by phase contrast. Scale bar, 100 μM.

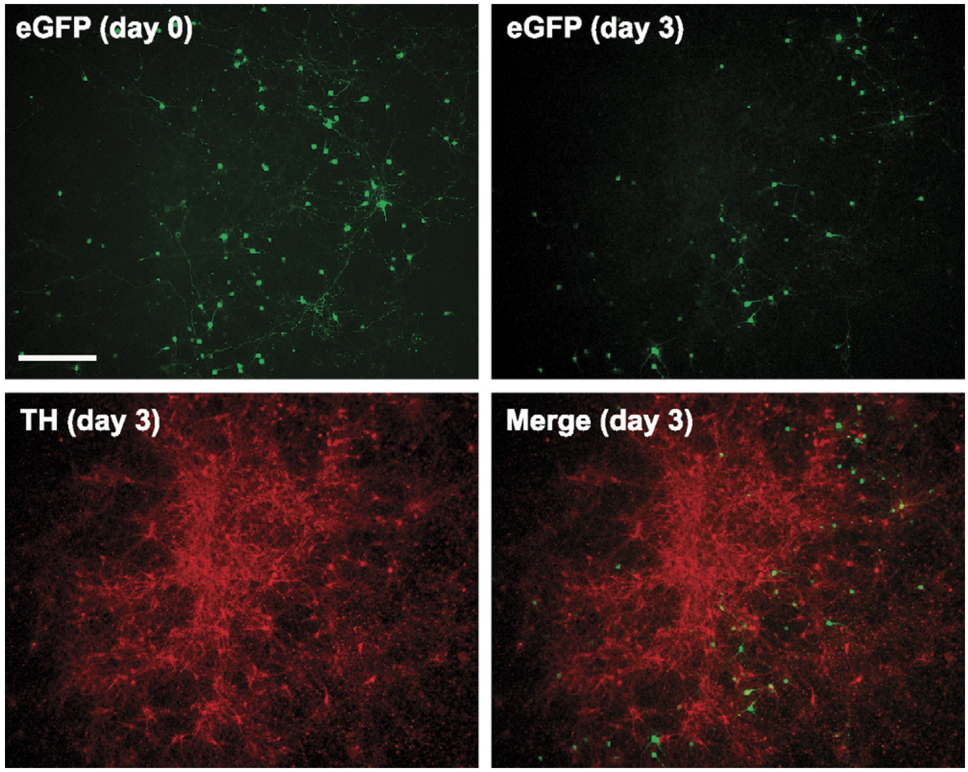
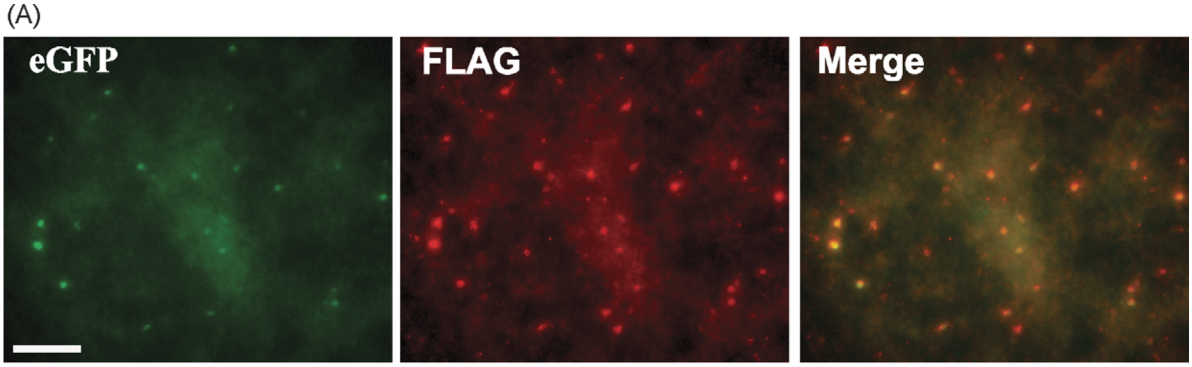


Fig. 2. eGFP-transfected TH-positive neurons die by GDNF deprivation. A typical pcDNA3.1/eGFP-co-transfected mesencephalic culture imaged just after GDNF deprivation (Day 0) and at the third day of GDNF deprivation after staining with anti-TH antibodies. Note the reduction of eGFP-TH-double-positive neurons by GDNF deprivation. The remaining eGFP-positive (and TH-negative) neurons are subtracted from the counted results. Scale bar, 100 μM.

(A)



(B)

GFRα1-FLAG	95.2% ± 1.0; 334 neurons
XIAP-myc	92.3% ± 2.2; 325 neurons

Fig. 3. Almost all neurons expressing eGFP reporter also express another co-transfected plasmid. The 6 DIV GDNF-dependent neurons were transfected with GFRα1-FLAG/eGFP or XIAP-myc/eGFP (both at ratio 5:1), grown for three days with GDNF, then fixed and stained with anti-FLAG or anti-myc antibody respectively. (A) Example of eGFP positive, anti-FLAG positive neurons and their co-localization. Scale bar, 100 μM. (B) Quantitation of neurons co-expressing eGFP and FLAG or myc immunoreactivity, calculated as a percentage of all eGFP-positive neurons. The means ± S.E.M. of three independent experiments, as well as the averaged absolute numbers of counted neurons are shown.

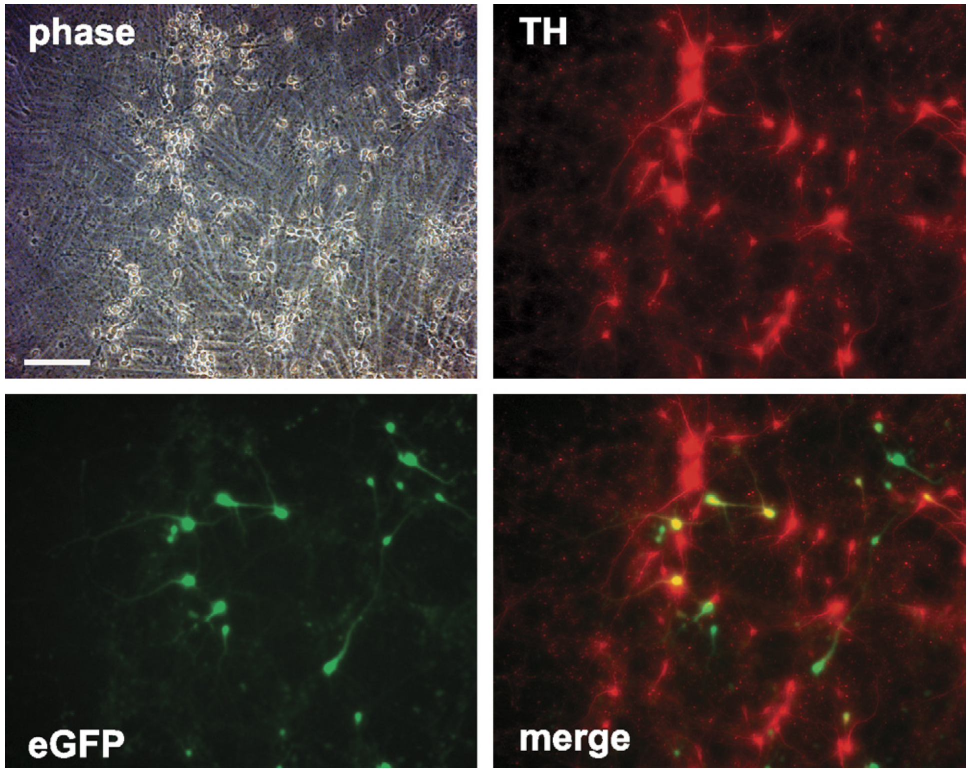


Fig. 4. Transfected eGFP is expressed in both TH-positive and TH-negative neurons. The neurons were maintained with GDNF for 6 DIV, transfected with expression plasmid for eGFP, fixed the next day and immunostained for TH. Note in merging of eGFP- and TH-stained images, the yellow neurons are the transfected dopaminergic neurons, whereas the green neurons are the transfected non-dopaminergic neurons. Scale bar, 100 μM.

age (ratio 1:5), the vast majority of green (eGFP-expressing) neurons were also red (immunopositive for FLAG or Myc) (Fig. 3). Thus, in such a ratio of plasmids, eGFP expression adequately represents the neurons expressing the plasmid of interest.

3.4. Counting the transfected neurons

In our assay, we overexpressed anti-apoptotic proteins in the apoptotic neurons and measured the effect by increased survival of the neurons, as compared to mock-transfected cultures. The fraction of transfected DA neurons was always too low to allow in batch measurement of survival. For example, live–dead kits or measurements of total TH immunofluorescence were not applicable (not shown). Therefore, the only choice to get accurate data was to count the living transfected neurons "manually" under a microscope. Counting of all eGFP-positive neurons from each dish was, however, technically feasible as the neurons were plated onto sufficiently small areas (Fig. 2).

Monitoring the survival of transfected (eGFP-expressing) neurons after GDNF deprivation appeared problematic. First, the survival of the neurons on the third day of GDNF deprivation must be normalized to the number of initial neurons living at the time of GDNF deprivation. We noticed that the transfection efficiency often varies between the dishes of the same culture. Therefore, it was essential to count the initial and final number of transfected neurons from the same individual dish, rather than relying on separate dishes for normalization. Usage of co-transfected eGFP allows us to count the transfected neurons twice from each dish: immediately at, and three days after GDNF deprivation.

Second, both dopaminergic and non-dopaminergic neurons become transfected in the mixed culture, obviously in a random manner. Fig. 4 illustrates a typical situation where some transfected (eGFP-expressing) neurons also expressed TH (the neurons relevant for the experiment) and some did not (the non-relevant neurons) but both were counted in our assay. To discount the non-relevant neurons, we stained the cultures with TH antibodies at the end of the experiment, visualizing the immunostaining with red-emitting fluorochrome (Cy3). We then counted separately the neurons co-expressing eGFP and TH (the final number of transfected neurons), and the neurons expressing eGFP but not TH (the non-relevant neurons). The latter numbers were subtracted from the respective initial counts. This approach was relevant for our GDNF deprivation studies, as the TH-negative non-dopaminergic neurons did not die due to GDNF deprivation but, instead, persisted throughout the experiment. Fig. 2 shows a representative culture co-transfected with eGFP and mock plasmid (Day 0), deprived of GDNF for three days and then stained with anti-TH antibodies. Note the very small number of eGFP-TH-double positive neurons (most of them died due to GDNF removal), whereas there are still many neurons expressing eGFP alone (persisting due to insensitivity to GDNF).

Table 1
Counting of transfected TH-positive neurons by co-transfected eGFP in the GDNF deprivation assay

	DN-casp-6, GDNF+	DN-casp-6, GDNF−	Vector, GDNF+	Vector, GDNF−
Day 0, eGFP+	195 ± 41	244 ± 33	163 ± 24	193 ± 42
Day 3, eGFP+/TH+	94 ± 15	81 ± 13	95 ± 14	24 ± 4
Day 3, eGFP+/TH−	78 ± 15	93 ± 18	48 ± 8	70 ± 18

6 DIV GDNF-maintained midbrain cultures transfected with dominant-negative caspase-6 (DN-casp-6) or empty vector (pcDNA3.1), counted the next day after transfection (and immediately after GDNF deprivation) (Day 0), and three days after GDNF deprivation (Day 3) after staining with TH antibodies. The averaged absolute numbers (± S.E.M; $n = 4$) of counted neurons are shown.

3.5. Blocking of caspase-6 inhibits the death of GDNF-deprived dopaminergic neurons

As an example of the application of established technique, we transfected the 6 DIV GDNF-dependent cultures with the dominantly blocking isoform of caspase-6 that significantly inhibited the apoptosis of NGF-deprived sympathetic neurons (Yu et al., 2003). The sister cultures were mock-transfected with the empty vector pcDNA3. eGFP was always co-transfected at a ratio of 1:5 as explained above. The next day, GDNF was removed from the successfully transfected cultures and the initial number of eGFP-positive neurons counted from each dish. Three days later the cultures were immunostained for TH and the TH-positive/eGFP-positive neurons (final number) counted from each dish. The TH-negative/eGFP-positive neurons were also counted and subtracted from the corresponding initial numbers. The final number of neurons was normalized to the corrected initial numbers for each individual dish (% survival). As shown in Fig. 5, overexpression of dominantly blocking caspase-6 significantly inhibited the death of GDNF-deprived neurons, whereas the empty vector had no effect. The general toxicity of the procedure was low as the majority of the transfected DA neurons survived in the presence of GDNF (Fig. 5). For illustration, the averaged absolute numbers of neurons counted in this experiment (four repeats on independent cultures, each experimental group always in three parallels) are shown in Table 1. To get the initial number of transfected DA neurons, the non-DA neurons (row 3) were subtracted from the eGFP-positive neurons at Day 0 (row 1). These corrected initial numbers were then used to normalize the final counts (row 2). The numbers shown in Table 1 were typical for our transfection–survival assays.

We verified activation of caspase-6 in the GDNF-deprived DA neurons by immunoblotting with antibodies that recognize the large subunit of cleaved caspase-6. As shown in Fig. 5C, a band of expected size (about 18 kDa) was found in the GDNF-deprived, but not GDNF-maintained cultures.

4. Discussion

In this paper we developed a new method to study the role of death pathways in apoptotic DA neurons using transiently transfected dissociated midbrain cultures.

To get reliable results, the culture conditions appeared to be critical (Takeshima et al., 1996). We introduced several technical aspects to that end. First, to get the experimental groups comparable, we plated the neurons on microislands of fixed standard size, so that the number of the neurons did not vary much between the dishes of the given assay. The microislands were small enough to count all transfected neurons "manually" under the microscope. Second, we cultured the neurons with GDNF for 5–6 days before starting the experiment (transfection and GDNF deprivation). During these days, the neurons recovered from the shock of tissue dissociation and axotomy. Also the number of TH-positive neurons in the culture stabilized under the influence of GDNF during these days. Removal of GDNF from the 5–6-DIV cultures triggered the death of DA neurons that solely depends on the absence of neurotrophic support and could be blocked by caspase inhibitors (not shown). Also, serum was used only briefly during the initial preparation to prime the neurons with TGF-β (Krieglstein et al., 1996). Thereafter, the cultures were grown without serum to exclude any serum-derived neurotrophic influence.

Transfection of DA neurons appeared to be a difficult task. We tried several commercially available transfection techniques and found them not to be satisfactory, as the efficiency was too low or the procedure was toxic for the neurons. For example, Lipofectamine or Effectene produced too low number of transfected neurons that only rarely were TH-positive. Neuroporter, reported to transfect efficiently the primary hippocampal neurons, transfected only a specific subpopulation of the neurons in our cultures that were strictly TH-negative. Actually, most of the current neuronal transfection techniques, established mainly on the hippocampal neurons, did not work well on the midbrain cultures. We found empirically that calcium phosphate co-precipitation procedure gives satisfactory results, but there could still be more efficient techniques. We were not able to achieve standard transfection efficiency, and sometimes the procedure was toxic to the culture due to unknown reasons.

Usage of co-transfected eGFP provided several advantages to our transfection–survival assay. First, it allows identification of the transfected cells with certainty. When the plasmid of interest was applied in excess, virtually all eGFP-expressing cells co-expressed it. Second, it allows counting of the transfected neurons from the same living cultures to be done several times, which is not possible by immunostaining that requires fixation. Thus, the final numbers of transfected neurons were always normalized to the initial numbers on the same dish, not on the separate normalizing dish, that made our transfection–survival assay quantitatively reliable.

The major problem in our assay was that the midbrain cultures contain large fraction of non-DA neurons that do not die by GDNF deprivation. Still these neurons were transfected and

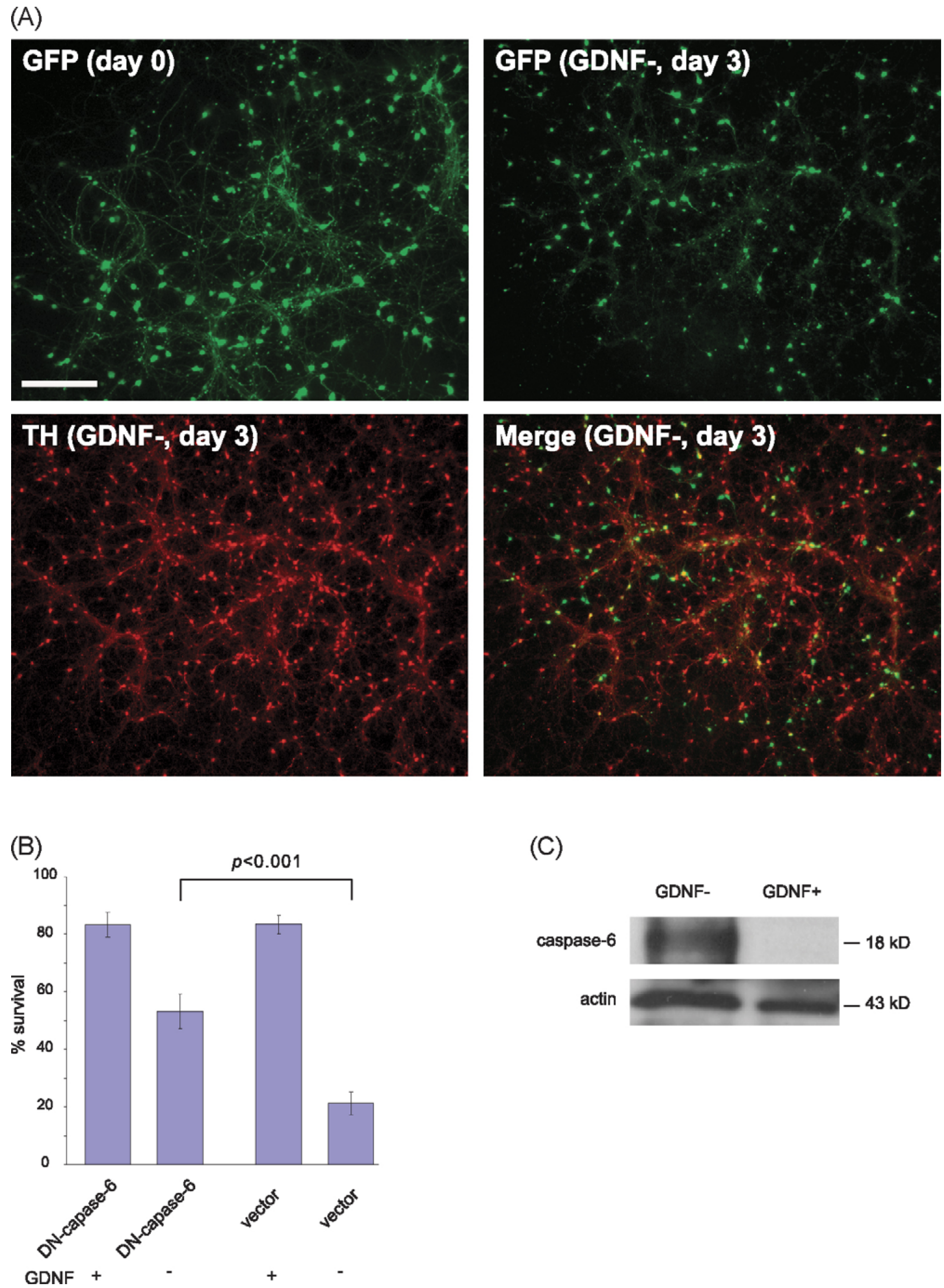


Fig. 5. Transfected DN-caspase-6 prevents death of TH-positive neurons. (A) Typical example of DN-caspase-6/eGFP-transfected mesencephalic culture just at (Day 0) or three days after (Day 3) GDNF deprivation, immunostained for TH. Note that many eGFP-expressing, TH-positive (yellow on the merged image) neurons survived (compare to vector/eGFP-transfected control cultures, Fig. 2). Scale bar, 100 μM. (B) Survival of the GDNF-deprived or GDNF-maintained DA neurons overexpressing DN-caspase-6 or empty vector. The means ± S.E.M. of three independent experiments are shown. The means were compared by Student's *t*-test; null hypothesis was rejected at a significance level of 0.05. (C) Cleavage product of caspase-6 is found in the GDNF-deprived (GDNF-) but not GDNF-maintained (GDNF+) cultures (upper panel). Lower panel shows reprobing of the filter with antibodies to actin.

counted, this distorted the results. We initially tried to make use of transgenic mice with eGFP in the TH-expressing cells, thus having the DA neurons (but not other midbrain neurons) fluorescent (Sawamoto et al., 2001). However, the eGFP fluorescence was lost during the 5–6-day culture. We then applied a post hoc subtraction technique where the eGFP-positive, TH-negative neurons were counted for each dish at the end of the experiment and subtracted from the total eGFP-counts. Indeed, the non-TH neurons did not show any response to GDNF deprivation throughout the experiment. Such a correction allows us to get true numbers for the transfected DA neurons.

It must be noted that our approach is not convenient for overexpressed pro-apoptotic proteins (e.g. Bax) that rapidly kills the transfected neurons, both dopaminergic and non-dopaminergic (not shown). In such conditions, following the fate of transfected neurons appears impossible. Usage of the inducible promoter in the plasmid for a pro-apoptotic protein would make it possible to follow the initial and final numbers of all transfected neurons. Still, it is not possible to know the number of transfected non-TH neurons. However, our approach is valuable in studies of anti-apoptotic effects, either overepression of anti-apoptotic proteins or suppression of pro-apoptotic ones. The latter case was exemplified here by blocking pro-apoptotic caspase-6 that potently inhibited the death of GDNF-deprived neurons. Good reproducibility of the results on independent cultures shows the reliability of our assay.

The caspases that are activated by GDNF deprivation (or, conversely, suppressed by GDNF) in the DA neurons are very poorly known. Here we show that caspase-6, an executionary caspase, is clearly involved in this death. Our technique can now be used to study the role of other components of the death pathway, including apical caspases that cleave and activate caspase-6.

Acknowledgments

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